

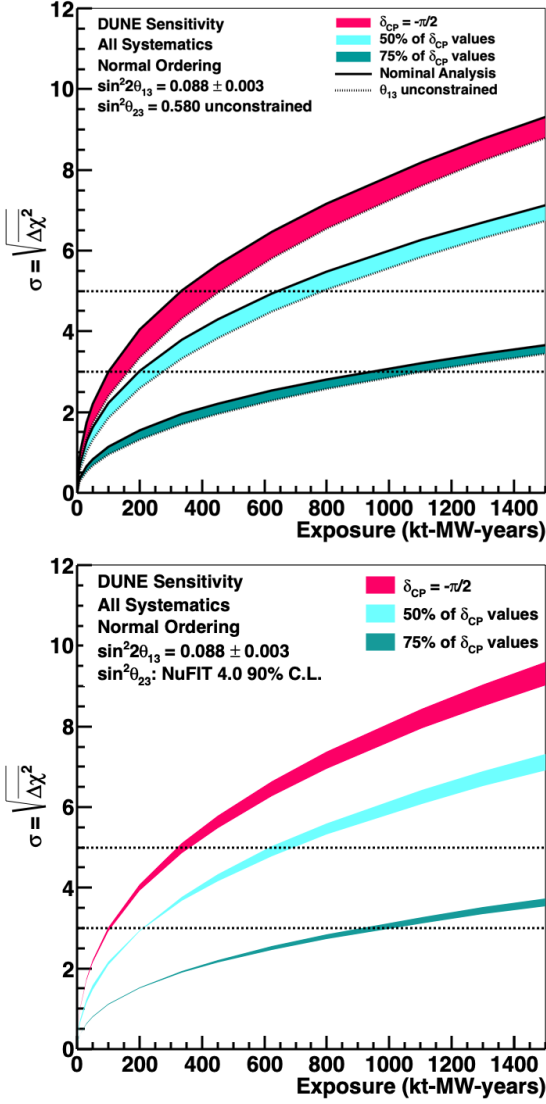


Fig. 16 Significance of the DUNE determination of CP-violation ($\delta_{CP} \neq [0, \pi]$) for the case when $\delta_{CP} = -\pi/2$, and for 50% and 75% of possible true δ_{CP} values, as a function of exposure in kt-MW-years. Top: The width of the band shows the impact of applying an external constraint on θ_{13} . Bottom: The width of the band shows the impact of varying the true value of $\sin^2 \theta_{23}$ within the NuFIT 4.0 90% C.L. region.

$\delta_{CP} = -\pi/2$, as a function of exposure in kt-MW-years, which can be converted to years using the staging scenario described in Section 6. The width of the bands show the impact of applying an external constraint on θ_{13} . CP violation can be observed with 5σ significance after about seven years (336 kt-MW-years) if $\delta_{CP} = -\pi/2$ and after about ten years (624 kt-MW-years) for 50% of δ_{CP} values. CP violation can be observed with 3σ significance for 75% of δ_{CP} values after about 13 years of running. In the bottom plot of Figure 16, the width of the bands shows the impact of applying an external constraint on θ_{13} , while in the bottom plot,

the width of the bands is the result of varying the true value of $\sin^2 \theta_{23}$ within the NuFIT 4.0 90% C.L. allowed region.

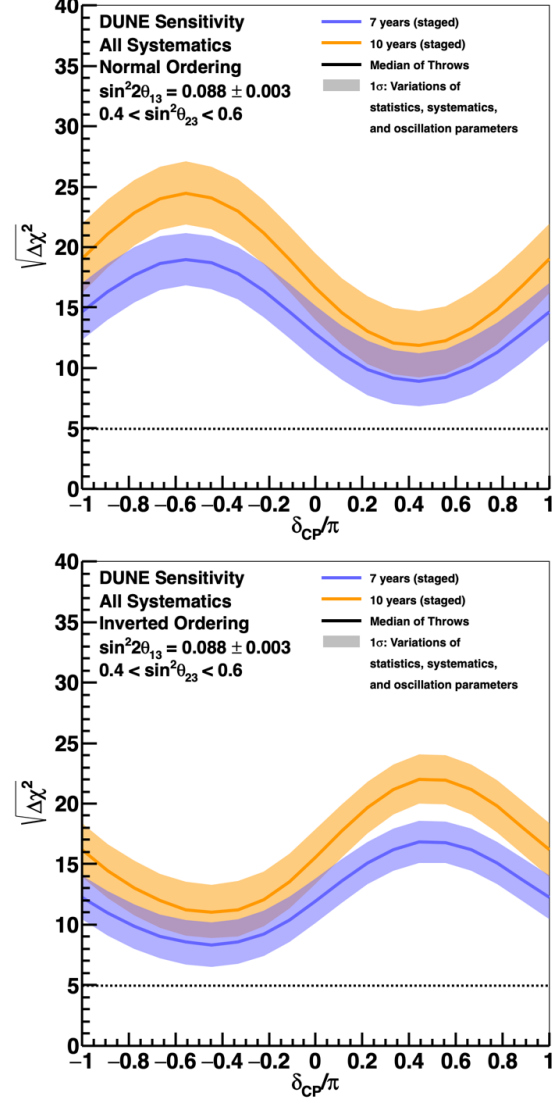


Fig. 17 Significance of the DUNE determination of the neutrino mass ordering, as a function of the true value of δ_{CP} , for seven (blue) and ten (orange) years of exposure. The width of the transparent bands cover 68% of fits in which random throws are used to simulate statistical variations and select true values of the oscillation and systematic uncertainty parameters, constrained by pre-fit uncertainties. The solid lines show the median sensitivity.

Figure 17 shows the significance with which the neutrino mass ordering can be determined in both NO and IO as a function of the true value of δ_{CP} , for both seven and ten year exposures, including the effect of all other oscillation and systematic parameters using the toy throwing method described in Section 8. The characteristic shape results from near degeneracy between